VCE PHYSICS · PRACTICE EXAM

Physics · Harder paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ **Date** _____

115 marks

reading 15 min · writing 150 min

State average on these questions

52 / 115 (45%)

a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright (some exams include third-party material).
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Methods and science working can be typed in plain text, e.g. v = sqrt(2gh) = 9.9 m/s.
3. Keep to the real time limits above, and don't look at the guide until you've finished.
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5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

Every question is a real VCAA exam question, chosen by ranking all the questions from the listed exams by how Victorian students actually scored on them (percentage correct for multiple choice, state average ÷ marks for written questions). This paper takes the hardest questions. Questions on content that isn't in the current study design have been left out.

Sources

- 2022 Physics exam (PDF) · examiners' report *(previous study design)*
- 2023 Physics exam (PDF) · examiners' report *(previous study design)*
- 2024 Physics exam (PDF) · examiners' report
- 2025 Physics exam (PDF) · examiners' report

Section A – Multiple-choice questions

Open each question in the official exam PDF (click the blue link), then select A–D. One mark each.

Question 1 · 1 mark

2022 VCAA exam · Section A · Question 4 · previous study design · open in the official exam PDF (p. 4) ↗

Where the field between Q and 4Q is zero

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 2 · 1 mark

2022 VCAA exam · Section A · Question 7 · previous study design · open in the official exam PDF (p. 6) ↗

Comparing collision forces between two trucks

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 3 · 1 mark

2022 VCAA exam · Section A · Question 9 · previous study design · open in the official exam PDF (p. 7) ↗

Force of a rope on each student in a tug of war

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 4 · 1 mark

2023 VCAA exam · Section A · Question 3 · previous study design · open in the official exam PDF (p. 3) ↗

Altitude where g is half its surface value

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 5 · 1 mark

2023 VCAA exam · Section A · Question 4 · previous study design · open in the official exam PDF (p. 4) ↗

Impulse from a force–time graph

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 6 · 1 mark

2023 VCAA exam · Section A · Question 11 · previous study design · open in the official exam PDF (p. 8) ↗

Extra energy stored when a spring compresses further

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 7 · 1 mark

2024 VCAA exam · Section A · Question 2 · open in the official exam PDF (p. 2) ↗

Orbital radius after a satellite's mass doubles

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 8 · 1 mark

2024 VCAA exam · Section A · Question 3 · open in the official exam PDF (p. 3) ↗

Forces on a ball moving in a horizontal circle on a string

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 9 · 1 mark

2024 VCAA exam · Section A · Question 7 · open in the official exam PDF (p. 7) ↗

Force and torque on a loop rotated through 90°

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 10 · 1 mark

2024 VCAA exam · Section A · Question 9 · open in the official exam PDF (p. 8) ↗

New separation when the force between charges becomes 6F

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 11 · 1 mark

2024 VCAA exam · Section A · Question 10 · open in the official exam PDF (p. 9) ↗

Choosing the flux–time graph for a given induced current

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 12 · 1 mark

2024 VCAA exam · Section A · Question 12 · open in the official exam PDF (p. 10) ↗

Induced current as a coil moves away from a magnet

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 13 · 1 mark

2024 VCAA exam · Section A · Question 13 · open in the official exam PDF (p. 11) ↗

Output of a generator with a split-ring commutator

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 14 · 1 mark

2024 VCAA exam · Section A · Question 15 · open in the official exam PDF (p. 12) ↗

Effect of a booster ring on electrons already near c

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 15 · 1 mark

2024 VCAA exam · Section A · Question 16 · open in the official exam PDF (p. 12) ↗

Matching energy-level transitions to spectral lines

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 16 · 1 mark

2025 VCAA exam · Section A · Question 3 · open in the official exam PDF (p. 4) ↗

Speed of a puck after a head-on collision

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 17 · 1 mark

2025 VCAA exam · Section A · Question 8 · open in the official exam PDF (p. 8) ↗

Where the field between +Q and +9Q is zero

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 18 · 1 mark

2025 VCAA exam · Section A · Question 9 · open in the official exam PDF (p. 9) ↗

Work done on a satellite in one circular orbit

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 19 · 1 mark

2025 VCAA exam · Section A · Question 11 · open in the official exam PDF (p. 10) ↗

Comparing a globe on 6 V AC RMS and on 6 V DC

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 20 · 1 mark

2025 VCAA exam · Section A · Question 20 · open in the official exam PDF (p. 15) ↗

Linearising speed against distance fallen

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Section B – Short-answer questions

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Question 1 (6 marks)

2022 VCAA exam · Section B · Question 4 · previous study design · open in the official exam PDF (p. 18) ↗

Square loop moved near a long current-carrying wire

EMF–time sketch and current direction for each of three movements (6 marks)

Question 2 (2 marks)

2022 VCAA exam · Section B · Question 11 · previous study design · open in the official exam PDF (p. 31) ↗

Why atmospheric muons reach Earth’s surface

Explain using special relativity (2 marks)

Question 3 (6 marks)

2022 VCAA exam · Section B · Question 12 · previous study design · open in the official exam PDF (p. 32) ↗

Double-slit bands: bright centre, wave evidence, band spacing

a. Why C is bright and the band beside it is dark (2 marks)

b. How the experiment supported the wave model of light (2 marks)

c. Spacing of the dark bands (2 marks)

Question 4 (5 marks)

2023 VCAA exam · Section B · Question 3 · previous study design · open in the official exam PDF (p. 15) ↗

Two parallel wires with opposite currents

a. Sketch the magnetic field around the wires (≥ 5 lines) (3 marks)

b. Do the wires attract or repel? Explain (2 marks)

Question 5 (7 marks)

2023 VCAA exam · Section B · Question 9 · previous study design · open in the official exam PDF (p. 26) ↗

Tennis serve: toss time, racquet speed, clearing the net

a. Show the ball is in the air for 1.2 s (2 marks)

b. Speed of the racquet head at point C (2 marks)

c. How far above the net the ball passes (3 marks)

Question 6 (6 marks)

2023 VCAA exam · Section B · Question 10 · previous study design · open in the official exam PDF (p. 29) ↗

Proton in an accelerator with γ = 2.00

a. Speed of the proton as a fraction of c (3 sig. figs) (3 marks)

b. Length of the beamline in the proton's frame (1 mark)

c. Kinetic energy of the proton in joules (2 marks)

Question 7 (3 marks)

2024 VCAA exam · Section B · Question 1 · open in the official exam PDF (p. 16) ↗

Tug boat towing two boats: acceleration and rope tension

a. Show the acceleration is 6.0×10^{-2} m s^{-2} (1 mark)

b. Tension in Rope 2 (2 marks)

Question 8 (5 marks)

2024 VCAA exam · Section B · Question 9 · open in the official exam PDF (p. 30) ↗

Simple DC motor: zero force on a side, non-starting orientation, more torque

a. An orientation where the force on side FG is zero, justified (2 marks)

b. A starting orientation where the coil won't start rotating, justified (2 marks)

c. One way to increase the torque with the same coil and battery (1 mark)

Question 9 (3 marks)

2024 VCAA exam · Section B · Question 14 · open in the official exam PDF (p. 40) ↗

PET scanner: electron–positron annihilation

a. Total energy released in the annihilation (2 marks)

b. Why the two gamma rays travel in opposite directions (1 mark)

Question 10 (20 marks)

2024 VCAA exam · Section B · Question 16 · open in the official exam PDF (p. 43) ↗

Photoelectric investigation (20 marks): variables, graph, analysis, filters

a. Controlled, dependent and independent variables (3 marks)

b. Plot the data with uncertainty bars and a line of best fit (3 marks)

c.i. Planck's constant from your graph (2 marks)

c.ii. Threshold frequency from your graph (1 mark)

c.iii. Work function from your graph (1 mark)

d. What the stopping voltages tell you about the photoelectrons (2 marks)

e. Effect of a different metal on the graph (1 mark)

f. Does P move with blue light at the same power? Justify (2 marks)

g. Is the saturation current NM greater, the same or lower with blue light? (2 marks)

h. One feature of the results that is evidence for particle-like light (3 marks)

Question 11 (6 marks)

2025 VCAA exam · Section B · Question 2 · open in the official exam PDF (p. 18) ↗

Airbag crash test: impulse, average force, force–displacement graph

a. Impulse the head exerts on the airbag (1 mark)

b. Why airbags help: compare average forces (2 marks)

c. Maximum force from the force–displacement graph (3 marks)

Question 12 (4 marks)

2025 VCAA exam · Section B · Question 5 · open in the official exam PDF (p. 24) ↗

Electric field from two point charges at a square's corner

a. Show the field at D due to A is 1.08×10^{7} N C^{-1} (1 mark)

b. Net field at D (magnitude and direction) (3 marks)

Question 13 (4 marks)

2025 VCAA exam · Section B · Question 6 · open in the official exam PDF (p. 26) ↗

Electrons accelerated between parallel plates

a. Electric force on the electron between the plates (2 marks)

b. Effect on kinetic energy gained when the plate gap doubles (2 marks)

Question 14 (5 marks)

2025 VCAA exam · Section B · Question 10 · open in the official exam PDF (p. 30) ↗

Magnet falling through a solenoid: induced EMF spikes

a. Explain why a current is observed (2 marks)

b. Why spike B is opposite in sign to A and larger (3 marks)

Question 15 (7 marks)

2025 VCAA exam · Section B · Question 11 · open in the official exam PDF (p. 32) ↗

Wind turbine powering heat lamps: lamp count, transformers

a. Show the lamp current is 0.33 A (1 mark)

b. How many lamps are installed? Show working (3 marks)

c.i. Type of transformer for T1 and T2 (1 mark)

c.ii. Justify the transformer choices (2 marks)

Question 16 (3 marks)

2025 VCAA exam · Section B · Question 13 · open in the official exam PDF (p. 36) ↗

Accelerating voltage for electrons of a given de Broglie wavelength

Find the accelerating voltage in kV (3 marks)

Question 17 (3 marks)

2025 VCAA exam · Section B · Question 14 · open in the official exam PDF (p. 37) ↗

Michelson–Morley null result and special relativity

Explain how the null result supports Einstein’s theory (3 marks)

Marking guide follows

Marking guide

Mark yourself only after you've finished. Tick the boxes you earned, enter your marks and compare with how Victoria went on each question.

Section A – Multiple-choice questions

✓	Q	Ans	Why	How Victoria went
☐	1	**D**	$kQ/x^2 = 4kQ/(12 - x)^2$: $(12 - x)/x = 2$, so $x = 4\ \text{cm}$ from Q. Only 18% got this.	18% correct A 26% · B 31% · C 25% · D 18%
☐	2	**C**	Newton's third law: equal in size and opposite in direction, $F_{X \text{ on } Y} = -F_{Y \text{ on } X}$, whatever the masses.	47% correct A 5% · B 19% · C 47% · D 29%
☐	3	**B**	Newton's third law: the rope pulls on each student with the same 400 N that each student applies. The forces are not added.	52% correct A 33% · B 52% · C 2% · D 13%
☐	4	**C**	$g \propto 1/r^2$: half of g at $r = \sqrt2 R$. Altitude $= \sqrt2R - R$. Don't forget to subtract R.	39% correct A 25% · B 24% · C 39% · D 11%
☐	5	**B**	Impulse = area under the F–t graph. Count the squares (about 10) × the value of each square ≈ 5 N s.	49% correct A 4% · B 49% · C 33% · D 13%
☐	6	**B**	$\Delta E = \tfrac12k(x_2^2 - x_1^2)$ with $k = 5\times10^6$: 160 J − 40 J = 120 J. Not $\tfrac12k(4\ \text{mm})^2$.	31% correct A 36% · B 31% · C 25% · D 7%
☐	7	**B**	$v = \sqrt{GM/r}$ doesn't involve the satellite's mass, so with the same speed the radius stays R.	39% correct A 12% · B 39% · C 39% · D 10%
☐	8	**A**	Only two forces act: weight (down) and tension (along the string). Their resultant provides the centripetal force. There is no separate "centripetal force".	51% correct A 51% · B 4% · C 37% · D 8%
☐	9	**D**	The force on PQ is unchanged (same current, same field, still perpendicular). The torque drops, because the force now acts parallel to the plane of the loop.	39% correct A 12% · B 24% · C 25% · D 39%
☐	10	**B**	$F \propto 1/d^2$, so $d_{new} = d/\sqrt6 \approx 0.41d$.	54% correct A 25% · B 54% · C 13% · D 8%
☐	11	**C**	Induced current ∝ the gradient of the flux–time graph. A constant current means a straight-line flux graph, and a bigger current means a steeper segment.	34% correct A 9% · B 22% · C 34% · D 35%
☐	12	**C**	Moving up from the N pole, the upward flux decreases. The induced current opposes this by making an upward field inside the coil, which needs an anticlockwise current viewed from above.	42% correct A 30% · B 11% · C 42% · D 16%
☐	13	**A**	The commutator reverses the connection every half turn: a varying voltage that never changes direction, which is DC.	51% correct A 51% · B 11% · C 6% · D 32%
☐	14	**B**	At $\gamma = 200$ the speed is already almost c, so it barely changes. Energy $= \gamma mc^2$ rises in proportion to γ, which is a large increase.	47% correct A 31% · B 47% · C 17% · D 5%
☐	15	**D**	Larger energy gaps give higher-frequency lines ($E = hf$). Three big drops to E_1 and two small drops give three high-f lines and two low-f lines.	43% correct A 13% · B 27% · C 17% · D 43%
☐	16	**C**	Conserve momentum with a sign convention (right positive): $mu_1 - mu_2 = -mv_1 + mv_2$, so $v_2 = u_1 - u_2 + v_1$.	27% correct A 14% · B 7% · C 27% · D 51%
☐	17	**C**	Set $kQ/x^2 = 9kQ/(0.20 - x)^2$: $(0.20 - x)/x = 3$, so $x = 5.0\ \text{cm}$ from +Q.	33% correct A 16% · B 39% · C 33% · D 12%

☐	18	**B**	Gravity acts at 90° to the velocity, so no work is done and the kinetic energy stays constant.	36% correct A 11% · B 36% · C 18% · D 34%
☐	19	**B**	RMS is defined as the AC value that delivers the same average power as that DC voltage.	30% correct A 17% · B 30% · C 33% · D 19%
☐	20	**B**	From rest, $v^2 = 2gs$, so v^2 against s is a straight line through the origin (gradient $2g$).	44% correct A 21% · B 44% · C 15% · D 19%

Multiple-choice score: ☐ **/ 20** **(state average 8.1)**

Section B – Short-answer questions

Question 1 (6 marks) · state average 2.6 / 6 (43%)

2022 VCAA exam · Section B · Question 4

EMF–time sketch and current direction for each of three movements

- ☐ Rotation about the x-axis: EMF alternates (sinusoidal-type)
- ☐ Rotation: current alternating
- ☐ Moving parallel to the wire: EMF zero (flux constant)
- ☐ Parallel movement: no current
- ☐ Moving away from the wire: non-zero EMF (decreasing flux)
- ☐ Moving away: clockwise current (opposes the decreasing flux into the page)

Answer / notes: Alternating, none, clockwise.

State average 2.6 / 6 (43%); 7% scored full marks. Mark distribution: 0: 15% · 1: 18% · 2: 19% · 3: 15% · 4: 14% · 5: 12% · 6: 7%

Your mark: ☐ **/ 6**

Question 2 (2 marks) · state average 0.8 / 2 (40%)

2022 VCAA exam · Section B · Question 11

Explain using special relativity

- ☐ Time dilation (or, in the muon's frame, length contraction of the atmosphere)
- ☐ In Earth's frame the muons' half-life is longer than their proper half-life ($t = \gamma t_0$), so they travel much further before decaying

Answer / notes: Time dilation lets them survive the trip.

State average 0.8 / 2 (40%); 33% scored full marks. Mark distribution: 0: 52% · 1: 15% · 2: 33%

Your mark: ☐ **/ 2**

Question 3 (6 marks) · state average 3 / 6 (50%)

2022 VCAA exam · Section B · Question 12

a. Why C is bright and the band beside it is dark

- ☐ At C the path difference is zero, so constructive interference (bright)
- ☐ The first dark band has a path difference of ½λ, so destructive interference

Answer / notes: Zero versus half-wavelength path difference.

State average 0.6 / 2 (30%); 20% scored full marks. Mark distribution: 0: 61% · 1: 19% · 2: 20%

Your mark: ☐ **/ 2**

b. How the experiment supported the wave model of light

- [ ] It produced an interference pattern
- [ ] Interference is a wave phenomenon; a particle model predicts only two bands

Answer / notes: Interference.

State average 1.3 / 2 (65%); 55% scored full marks. Mark distribution: 0: 27% · 1: 17% · 2: 55%

Your mark: ____ **/ 2**

c. Spacing of the dark bands

- [ ] $\lambda = c/f = 5.0\times10^{-7}\ \text{m}$, and $\Delta x = \lambda L/d$
- [ ] $\Delta x = 0.010\ \text{m}$

Answer / notes: $\frac{5.0\times10^{-7} \times 2.00}{1.0\times10^{-4}} = 1.0\times10^{-2}\ \text{m}$

State average 1.1 / 2 (55%); 51% scored full marks. Mark distribution: 0: 46% · 1: 3% · 2: 51%

Your mark: ____ **/ 2**

Question 4 (5 marks) · state average 2.3 / 5 (46%)

2023 VCAA exam · Section B · Question 3

a. Sketch the magnetic field around the wires (≥ 5 lines)

- [ ] Circular field lines around each wire
- [ ] Correct and opposite directions of circulation (right-hand grip rule)
- [ ] At least five lines, with the fields reinforcing between the wires

Answer / notes: Concentric circles, circulating opposite ways.

State average 1.5 / 3 (50%); 28% scored full marks. Mark distribution: 0: 36% · 1: 14% · 2: 23% · 3: 28%

Your mark: ____ **/ 3**

b. Do the wires attract or repel? Explain

- [ ] Repel
- [ ] The field from one wire at the other, with the right-hand slap rule, gives a force away from it (opposite currents repel)

Answer / notes: They repel.

State average 0.8 / 2 (40%); 23% scored full marks. Mark distribution: 0: 49% · 1: 28% · 2: 23%

Your mark: ____ **/ 2**

Question 5 (7 marks) · state average 3.3 / 7 (47%)

2023 VCAA exam · Section B · Question 9

a. Show the ball is in the air for 1.2 s

- [ ] Launch speed from $v^2 = u^2 + 2as$: $u = \sqrt{2(9.8)(1.8)} = 5.9\ \text{m s}^{-1}$
- [ ] Time up $= u/g = 0.61\ \text{s}$, so the total is about 1.2 s

Answer / notes: Up and down are symmetric.

State average 1.4 / 2 (70%); 66% scored full marks. Mark distribution: 0: 32% · 1: 2% · 2: 66%

Your mark: ____ **/ 2**

b. Speed of the racquet head at point C

- [ ] Quarter of a circle: $\tfrac14 \times 2\pi(1.8) = 2.83\ \text{m}$ in the 0.61 s the ball takes to rise
- [ ] $v \approx 4.7\ \text{m s}^{-1}$

Answer / notes: About 4.7 m s^{-1}. Using the full circumference was the common error.

State average 0.5 / 2 (25%); 26% scored full marks. Mark distribution: 0: 74% · 1: 26%

Your mark: ____ **/ 2**

c. How far above the net the ball passes

- [ ] Time to the net: $12/24 = 0.50\ \text{s}$
- [ ] Drop in that time: $\tfrac12(9.8)(0.50)^2 = 1.23\ \text{m}$
- [ ] Height above net: $3.6 - 1.23 - 0.90 \approx 1.5\ \text{m}$

Answer / notes: About 1.5 m. Many stopped at the 1.225 m drop.

State average 1.4 / 3 (47%); 31% scored full marks. Mark distribution: 0: 40% · 1: 9% · 2: 20% · 3: 31%

Your mark: ____ **/ 3**

Question 6 (6 marks) · state average 2.7 / 6 (45%)

2023 VCAA exam · Section B · Question 10

a. Speed of the proton as a fraction of c (3 sig. figs)

- [ ] $\gamma = 1/\sqrt{1 - v^2/c^2}$ set up
- [ ] Rearranged: $v = c\sqrt{1 - 1/\gamma^2}$
- [ ] $v = 0.866c$

Answer / notes: 0.866c

State average 1.8 / 3 (60%); 43% scored full marks. Mark distribution: 0: 27% · 1: 12% · 2: 19% · 3: 43%

Your mark: ____ **/ 3**

b. Length of the beamline in the proton's frame

- [ ] $L = L_0/\gamma = 2.40\ \text{km}$

Answer / notes: 2.40 km

State average 0.8 / 1 (80%); 73% scored full marks. Mark distribution: 0: 27% · 1: 73%

Your mark: ____ **/ 1**

c. Kinetic energy of the proton in joules

- [ ] $E_k = (\gamma - 1)mc^2$
- [ ] $E_k \approx 1.50\times10^{-10}\ \text{J}$

Answer / notes: $(2.00 - 1)(1.67\times10^{-27})(3.0\times10^8)^2 = 1.50\times10^{-10}\ \text{J}$. Using ½mv² was the common error.

State average 0.1 / 2 (5%); 47% scored full marks. Mark distribution: 0: 49% · 1: 4% · 2: 47%

Your mark: ____ **/ 2**

Question 7 (3 marks) · state average 1.5 / 3 (50%)

2024 VCAA exam · Section B · Question 1

a. Show the acceleration is 6.0×10^{-2} m s^{-2}

- [ ] $a = \frac{9.2\times10^4 - 1.2\times10^4 - 1.4\times10^4}{5.0\times10^5 + 6.0\times10^5} = 6.0\times10^{-2}\ \text{m s}^{-2}$

Answer / notes: Only the two towed boats, with both resistances.

State average 0.6 / 1 (60%); 56% scored full marks. Mark distribution: 0: 44% · 1: 56%

Your mark: ____ **/ 1**

b. Tension in Rope 2

- [ ] For Boat 3: $T - 1.4\times10^4 = 6.0\times10^5 \times 0.060$
- [ ] $T = 5.0\times10^{4}\ \text{N}$

Answer / notes: $5.0\times10^4\ \text{N}$

State average 0.9 / 2 (45%); 41% scored full marks. Mark distribution: 0: 56% · 1: 3% · 2: 41%

Your mark: ____ **/ 2**

Question 8 (5 marks) · state average 2.5 / 5 (50%)

2024 VCAA exam · Section B · Question 9

a. An orientation where the force on side FG is zero, justified

- [ ] Coil horizontal (as shown)
- [ ] Current in FG is then parallel to the field, so F = nIlB gives zero

Answer / notes: Horizontal. Restating the orientation isn't a justification.

State average 0.9 / 2 (45%); 22% scored full marks. Mark distribution: 0: 33% · 1: 45% · 2: 22%

Your mark: ____ **/ 2**

b. A starting orientation where the coil won't start rotating, justified

- [ ] Coil vertical (plane perpendicular to the field)
- [ ] The forces on EF and GH act along the plane (no turning component), so the torque is zero

Answer / notes: Vertical.

State average 0.9 / 2 (45%); 29% scored full marks. Mark distribution: 0: 46% · 1: 24% · 2: 29%

Your mark: ____ **/ 2**

c. One way to increase the torque with the same coil and battery

- [ ] e.g. a stronger magnetic field (stronger magnets or a smaller gap)

Answer / notes: A stronger field.

State average 0.7 / 1 (70%); 65% scored full marks. Mark distribution: 0: 35% · 1: 65%

Your mark: ____ **/ 1**

Question 9 (3 marks) · state average 1 / 3 (33%)

2024 VCAA exam · Section B · Question 14

a. Total energy released in the annihilation

- [ ] $E = mc^2$ using the combined mass $2 \times 9.11\times10^{-31}\ \text{kg}$
- [ ] $E \approx 1.64\times10^{-13}\ \text{J}$

Answer / notes: $2 \times 9.11\times10^{-31} \times (3.0\times10^8)^2 = 1.64\times10^{-13}\ \text{J}$. Using one electron mass was the common error.

State average 0.8 / 2 (40%); 37% scored full marks. Mark distribution: 0: 60% · 1: 3% · 2: 37%

Your mark: ____ **/ 2**

b. Why the two gamma rays travel in opposite directions

- [ ] Conservation of momentum: the initial momentum is (almost) zero, so the photons must carry equal and opposite momentum

Answer / notes: Momentum conservation. Only 15% got this; most wrongly cited opposite charges.

State average 0.2 / 1 (20%); 15% scored full marks. Mark distribution: 0: 85% · 1: 15%

Your mark: ____ **/ 1**

Question 10 (20 marks) · state average 9.8 / 20 (49%)

2024 VCAA exam · Section B · Question 16

a. Controlled, dependent and independent variables

- [ ] Controlled: e.g. intensity/power of light, or the metal plate
- [ ] Dependent: stopping voltage
- [ ] Independent: frequency (filter)

Answer / notes: —

State average 2.0 / 3 (67%); 45% scored full marks. Mark distribution: 0: 16% · 1: 15% · 2: 24% · 3: 45%

Your mark: ____ **/ 3**

b. Plot the data with uncertainty bars and a line of best fit

- [ ] Points plotted correctly
- [ ] Uncertainty bars of ±0.1 V
- [ ] A straight line of best fit balanced through all points (not just first to last)

Answer / notes: The line should NOT simply join the first and last points.

State average 2.2 / 3 (73%); 47% scored full marks. Mark distribution: 0: 7% · 1: 13% · 2: 33% · 3: 47%

Your mark: ____ **/ 3**

c.i. Planck's constant from your graph

- [ ] Gradient calculated from two points on your line (rise/run)
- [ ] Value consistent with the line, about \(3.3\times10^{-15}\ \text{eV s}\) for this data

Answer / notes: Use the gradient of your line. Don't force it towards 4.14×10^{-15}.

State average 1.0 / 2 (50%); 42% scored full marks. Mark distribution: 0: 43% · 1: 15% · 2: 42%

Your mark: ____ **/ 2**

c.ii. Threshold frequency from your graph

- [ ] x-intercept, about \(5.5\times10^{14}\ \text{Hz}\) (consistent with your line)

Answer / notes: x-intercept

State average 0.7 / 1 (70%); 63% scored full marks. Mark distribution: 0: 37% · 1: 63%

Your mark: ____ **/ 1**

c.iii. Work function from your graph

- [ ] y-intercept magnitude, about 1.8 eV (consistent with your line)

Answer / notes: y-intercept

State average 0.5 / 1 (50%); 52% scored full marks. Mark distribution: 0: 48% · 1: 52%

Your mark: ____ **/ 1**

d. What the stopping voltages tell you about the photoelectrons

- [ ] They give the maximum kinetic energy of the emitted photoelectrons
- [ ] \(E_{k\,max} = qV_0\) (numerically equal in eV)

Answer / notes: The maximum KE of the electrons, not of the photons.

State average 0.5 / 2 (25%); 19% scored full marks. Mark distribution: 0: 72% · 1: 10% · 2: 19%

Your mark: ____ **/ 2**

e. Effect of a different metal on the graph

- [ ] A parallel line (same gradient, h) shifted, with different intercepts

Answer / notes: Same gradient, different intercepts.

State average 0.5 / 1 (50%); 44% scored full marks. Mark distribution: 0: 56% · 1: 44%

Your mark: ____ **/ 1**

f. Does P move with blue light at the same power? Justify

- [ ] P moves to the left (larger stopping voltage)
- [ ] Blue photons have more energy, so photoelectrons have greater maximum KE

Answer / notes: Left.

State average 0.9 / 2 (45%); 27% scored full marks. Mark distribution: 0: 38% · 1: 35% · 2: 27%

Your mark: ____ **/ 2**

g. Is the saturation current NM greater, the same or lower with blue light?

- [ ] Lower
- [ ] Same power but higher energy per photon means fewer photons per second, so fewer photoelectrons

Answer / notes: Lower. Only 3% got this.

State average 0.1 / 2 (5%); 3% scored full marks. Mark distribution: 0: 93% · 1: 4% · 2: 3%

Your mark: ____ **/ 2**

h. One feature of the results that is evidence for particle-like light

- [ ] A feature: threshold frequency, OR no time delay, OR Ek max depends on frequency not intensity
- [ ] What the wave model predicts for it (e.g. emission at any frequency if intense enough)
- [ ] How the photon model explains it (one photon, energy hf, per electron)

Answer / notes: —

State average 1.4 / 3 (47%); 29% scored full marks. Mark distribution: 0: 38% · 1: 14% · 2: 19% · 3: 29%

Your mark: ____ **/ 3**

Question 11 (6 marks) · state average 2.5 / 6 (42%)

2025 VCAA exam · Section B · Question 2

a. Impulse the head exerts on the airbag

- [ ] $|\Delta p| = mv = 5.0 \times 12 = 60\ \text{N s}$

Answer / notes: 60 N s

State average 0.9 / 1 (90%); 94% scored full marks. Mark distribution: 0: 6% · 1: 94%

Your mark: ____ **/ 1**

b. Why airbags help: compare average forces

- [ ] The change in momentum (impulse) is the same with or without the airbag
- [ ] The airbag increases the stopping time, so the average force $F = \Delta p/\Delta t$ is smaller

Answer / notes: Same impulse, longer time, smaller force.

State average 1.1 / 2 (55%); 45% scored full marks. Mark distribution: 0: 32% · 1: 23% · 2: 45%

Your mark: ____ **/ 2**

c. Maximum force from the force–displacement graph

- [ ] Work done = change in KE $= \tfrac12(5.0)(12)^2 = 360\ \text{J}$
- [ ] Area under the graph $= \tfrac12(0.20)F_{max} + 0.20F_{max} = 0.30F_{max}$
- [ ] $F_{max} = 1.2\times10^{3}\ \text{N}$

Answer / notes: $0.30F_{max} = 360$, so $F_{max} = 1200\ \text{N}$. Equating the area to the impulse (60) was the common error.

State average 0.5 / 3 (17%); 11% scored full marks. Mark distribution: 0: 71% · 1: 17% · 2: 11%

Your mark: ____ **/ 3**

Question 12 (4 marks) · state average 1.9 / 4 (48%)

2025 VCAA exam · Section B · Question 5

a. Show the field at D due to A is 1.08×10^{7} N C^{-1}

- [ ] $E = \frac{kq}{r^2} = \frac{8.99\times10^9 \times 3.00\times10^{-6}}{0.0500^2} = 1.08\times10^{7}\ \text{N C}^{-1}$

Answer / notes: Convert cm to m and square r.

State average 0.7 / 1 (70%); 67% scored full marks. Mark distribution: 0: 33% · 1: 67%

Your mark: ____ **/ 1**

b. Net field at D (magnitude and direction)

- [ ] The two fields at D are perpendicular (vector addition)
- [ ] $E = \sqrt2 \times 1.08\times10^7 \approx 1.53\times10^{7}\ \text{N C}^{-1}$
- [ ] Direction: south-west (away from both positive charges)

Answer / notes: $1.53\times10^7\ \text{N C}^{-1}$ south-west.

State average 1.2 / 3 (40%); 27% scored full marks. Mark distribution: 0: 45% · 1: 17% · 2: 11% · 3: 27%

Your mark: ☐ **/ 3**

Question 13 (4 marks) · state average 1.8 / 4 (45%)

2025 VCAA exam · Section B · Question 6

a. Electric force on the electron between the plates

- [ ] $E = V/d = 250/0.050 = 5.0\times10^{3}\ \text{V m}^{-1}$
- [ ] $F = qE = 8.0\times10^{-16}\ \text{N}$

Answer / notes: $8.0\times10^{-16}\ \text{N}$

State average 1.2 / 2 (60%); 57% scored full marks. Mark distribution: 0: 41% · 1: 2% · 2: 57%

Your mark: ☐ **/ 2**

b. Effect on kinetic energy gained when the plate gap doubles

- [ ] No change in the kinetic energy gained
- [ ] $\Delta E_k = qV$ depends only on the voltage (the force halves but the distance doubles)

Answer / notes: Unchanged. Most students predicted a change.

State average 0.6 / 2 (30%); 27% scored full marks. Mark distribution: 0: 68% · 1: 5% · 2: 27%

Your mark: ☐ **/ 2**

Question 14 (5 marks) · state average 2.4 / 5 (48%)

2025 VCAA exam · Section B · Question 10

a. Explain why a current is observed

- [ ] The moving magnet changes the magnetic flux through the solenoid
- [ ] By Faraday's law the changing flux induces an EMF, which drives a current in the circuit

Answer / notes: Changing flux, so EMF, so current.

State average 1.2 / 2 (60%); 57% scored full marks. Mark distribution: 0: 36% · 1: 7% · 2: 57%

Your mark: ☐ **/ 2**

b. Why spike B is opposite in sign to A and larger

- [ ] Entering: flux increasing; leaving: flux decreasing, so by Lenz's law the induced EMF reverses
- [ ] The magnet accelerates as it falls, so it is moving faster when it leaves
- [ ] A faster change of flux gives a larger EMF (Faraday's law)

Answer / notes: Opposite sign from Lenz; larger because the magnet is faster.

State average 1.2 / 3 (40%); 23% scored full marks. Mark distribution: 0: 44% · 1: 16% · 2: 17% · 3: 23%

Your mark: ☐ **/ 3**

Question 15 (7 marks) · state average 3 / 7 (43%)

2025 VCAA exam · Section B · Question 11

a. Show the lamp current is 0.33 A

- [ ] $I = P/V = 70/210 = 0.33\ \text{A}$

Answer / notes: —

State average 0.8 / 1 (80%); 84% scored full marks. Mark distribution: 0: 16% · 1: 84%

Your mark: ☐ **/ 1**

b. How many lamps are installed? Show working

- [ ] Voltage drop in the wires $= 230 - 210 = 20\ \text{V}$
- [ ] Line current $= 20/6.0 = 3.3\ \text{A}$
- [ ] Number of lamps $= 3.3/0.33 = 10$

Answer / notes: 10 lamps. Over 80% scored zero, because the voltage drop across the wires was the key.

State average 0.3 / 3 (10%); 9% scored full marks. Mark distribution: 0: 85% · 1: 4% · 2: 2% · 3: 9%

Your mark: ☐ **/ 3**

c.i. Type of transformer for T1 and T2

- [ ] T1 step-up and T2 step-down

Answer / notes: T1 step-up, T2 step-down.

State average 0.8 / 1 (80%); 81% scored full marks. Mark distribution: 0: 19% · 1: 81%

Your mark: ☐ **/ 1**

c.ii. Justify the transformer choices

- [ ] Step-up at the turbine lowers the transmission current, reducing I^2R losses
- [ ] Step-down at the shed returns the voltage to the level the lamps need

Answer / notes: —

State average 1.1 / 2 (55%); 42% scored full marks. Mark distribution: 0: 31% · 1: 27% · 2: 42%

Your mark: ☐ **/ 2**

Question 16 (3 marks) · state average 1.3 / 3 (43%)

2025 VCAA exam · Section B · Question 13

Find the accelerating voltage in kV

- [ ] $p = h/\lambda = 5.5\times10^{-23}\ \text{kg m s}^{-1}$
- [ ] $E_k = \frac{p^2}{2m} \approx 1.7\times10^{-15}\ \text{J}$ (or find v, then KE)
- [ ] $V = E_k/q \approx 10\ \text{kV}$ (10.5 kV)

Answer / notes: About 10.5 kV. Using E = pc (the photon formula) for electrons was a common error.

State average 1.3 / 3 (43%); 38% scored full marks. Mark distribution: 0: 49% · 1: 9% · 2: 4% · 3: 38%

Your mark: ☐ **/ 3**

Question 17 (3 marks) · state average 1.5 / 3 (50%)

2025 VCAA exam · Section B · Question 14

Explain how the null result supports Einstein's theory

- [ ] The experiment looked for a difference in the speed of light along perpendicular paths (motion through the aether)
- [ ] No difference was found, whatever the orientation: the null result
- [ ] This supports the postulate that the speed of light is the same for all observers (no aether is needed)

Answer / notes: —

State average 1.5 / 3 (50%); 28% scored full marks. Mark distribution: 0: 31% · 1: 19% · 2: 22% · 3: 28%

Your mark: ☐ **/ 3**

Your result

Total: _____ **/ 115** **State average on these questions: 52 / 115 (45%)**

Topics to revise:

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